Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Staff-level Software Engineer with 8 years of professional experience specializing in AI-enhanced product development, full-stack TypeScript architectures, and scalable cloud systems. Proven track record of taking end-to-end ownership of complex features, from designing multi-agent LLM pipelines and real-time voice/chat interfaces to building robust Node.js/Express and NestJS backend services deployed on GCP and AWS. Deep expertise in React, Next.js, and modern AI workflows (RAG, vector embeddings, prompt engineering), combined with a strong focus on developer experience, system design, and mentoring technical teams.

CORE COMPETENCIES

- **AI & LLM Systems:** Real-time voice interfaces, multimodality, RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), embeddings, prompt engineering, and agentic workflows.
- **Frontend Engineering:** React, Next.js, TypeScript, Tailwind CSS, Radix UI, state management (Redux, React Query, Zustand), and responsive, accessible UI design.
- **Backend & API Design:** Node.js, Express, NestJS, REST APIs, GraphQL, PostgreSQL, Prisma, MongoDB, JWT authentication, and RBAC.
- **Cloud & Infrastructure:** GCP, AWS, Docker, GitHub Actions, CI/CD pipelines, Nx monorepo orchestration, SSE streaming, and serverless architectures.
- **Testing & Quality:** Playwright, Jest, Cypress, ESLint, Prettier, Husky, Lighthouse, React Profiler, and AI-assisted development (Cursor, Claude Code, GitHub Copilot).
- **Leadership & Collaboration:** Technical architecture reviews, cross-functional alignment with Product and Design, mentoring engineers, Agile/Scrum, and driving organizational change.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer (AI & Fullstack Focus) *May 2023 - Present*

- Co-developed a full-stack **Next.js chat widget** powered by a **Retrieval-Augmented Generation (RAG) system** using **OpenAI's GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query data in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, migrating from a Vue.js web application to a **React Chrome extension**, optimizing workflows for clinical site integrations.
- Implemented authentication and authorization workflows on the frontend using **JWT-based flows**, securely handling access and refresh tokens with HTTP-only cookies, and enforcing permission-based UI rendering in collaboration with backend engineers.
- Architected a reusable design system in **React** and **Radix UI** that improved UI consistency, accessibility, and developer efficiency across products.
- Optimized application performance by implementing lazy loading, code splitting, and image optimization, significantly reducing load times as measured by **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, optimizing the build pipeline and refactoring legacy code to improve developer experience.
- Increased stakeholder confidence by writing end-to-end tests with **Playwright** in the **CI/CD pipeline**, publishing test results automatically on every push.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and knowledge sharing.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating closely with software architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance and load efficiency by identifying and resolving render bottlenecks using **React Profiler** and **Lighthouse Audits**, reducing bundle size by **30-40%**, which led to a **35% faster load time** and smoother runtime experience.
- Drove a **40% improvement** in React application performance by optimizing caching mechanisms, significantly boosting speed and user experience.
- Enhanced the development workflow by introducing automated testing frameworks, including **Jest** and **Enzyme**, which led to a **20% decrease** in bug reports.
- Established engineering standards for code quality, testing, and deployment across the team using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.

- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile project management practices.
- Mentored junior engineers and guided frontend best practices, fostering a culture of high-quality, maintainable code and consistent design.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Played a pivotal role in a team of 3 frontend engineers to build a high-volume bank Settlement Solution, significantly enhancing transaction processing capabilities to handle thousands to millions of transactions daily, boosting transaction settlement rates by **80%**.
- Spearheaded the development of critical modules for the Settlement Solution, including MasterCard, VISA, Express-cash, and E-commerce modules, using **React** with **Redux** for state management, **RxJS** for data streaming, and **Axios** for asynchronous API requests.
- Developed a real-time account reconciliation solution, boosting the reconciliation of daily transactions by **60%** through innovative transaction processing and data management strategies.
- Enhanced system performance and user experience by adopting the native browser **EventSource API** for real-time data streaming, replacing conventional socket.io methods for faster data updates.
- Built dynamic data visualization dashboards using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and Ant Design UI.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** to power new features, focusing on testable, maintainable, and well-documented code.
- Translated business requirements into efficient **MongoDB** database models and schemas, collaborating closely with the business team.
- Transformed user journeys and Figma designs into reusable **React** components, ensuring a consistent, intuitive user interface.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected and built the entire platform as an **Nx monorepo** managing 6 applications: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Designed and implemented a **multi-agent AI system** using the **Mastra** framework with 5 specialized agents for three-stage intent classification and domain-specific RAG-powered analysis.
- Built a **hybrid RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion, with **Cohere** reranking and **Voyage AI** embeddings (1024-dim, 708K+ chunks).
- Implemented **SSE streaming** for real-time chat responses with keepalive heartbeats, token-budgeted context management, conversation summarization, and memory extraction.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** end-to-end tests, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.
- Integrated **Langfuse** for LLM observability and scoring, **Infisical** for secrets management, and **Traefik** for reverse proxy routing.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science, 4.1 CGPA *Graduation Date: Sep 2019*